

Modeling of Electric Vehicle Loads for Power Flow Analysis based on PSAT

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Abstract—This paper proposes the modeling of electric vehicle (EV) loads for power flow analysis based on User Define Models (UDMs) in power system analysis toolbox (PSAT) programming. According to the development of EV technology especially motor controller and battery, the EV has totally designed and made for using in city area as small city cars. To study the performance of EV, this aims to present the model of EV loads for using to solve power flow analysis in the real power system. EV load is modeled as the function box equivalent on PSAT that can connect to the single line diagram of power system model. The IEEE 14 bus system is selected for simulating the performance of EV and charging station which compared between PQ load base case and EV load. The simulation result showed that the EV load was installed at bus No.14. The rated power of EV load was increased from 5.0 p.u. to 79 p.u. This proposed EV load model can be used to solve the power flow analysis with continuation power flow (CPF) method. The EV load model is directly effected to voltage stability margin when the EV load increased. This study can be verified that proposed EV load can use to study the EV load in the future works.

Keywords—Electric Vehicle Load; Power Flow Analysis; , Load Voltage Diaviation, Continuation Power Flow Method

I. INTRODUCTION

Recently, the study of electric vehicle (EV) in power system has been an interested to research on electrical power system. The high penetration of EV load in the electrical power system will be important to analyze and solve power flow for tracking power system problem. The energy management system (EMS) are required to system management and optimize of energy consumption from energy sources. Because of number of EV load charger may be charged at the same time at same power grid line. Therefore, it is necessary to develop an effective EV load and to investigate the impacts on power system operation.

Smart grid concept is the integration of electrical and communication infrastructures with advance process automation and information technologies within the existing electrical network as shown in Fig.1. The smart grid represents a complete change in the way utilities, politicians, customers, and other industry participants think about electricity delivery and its related services. Consequently, the role of EV load is coming to consume energy in demand side of grid and control

from EMS.

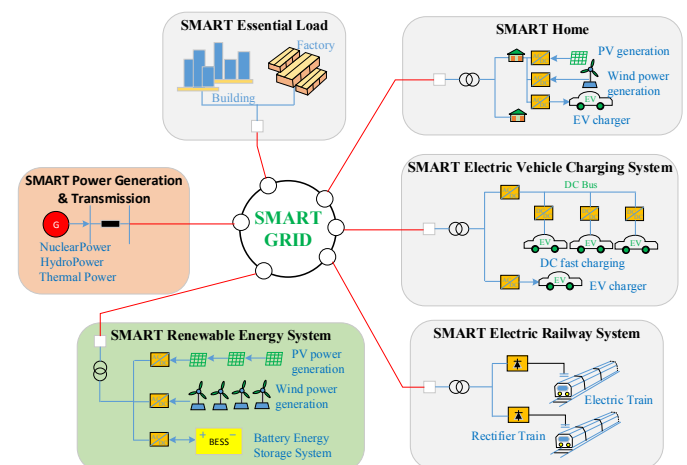


Fig.1 The SMART grid interfaced with Electric Vehicle load

The impact of EV charging station was effected to reduce the steady state voltage stability by the modeling based on charger of EV battery. It is observed that the power voltage relationship confirms the analytically derived load characteristics consist of constant power component and voltage dependent power component [1]. The EV load was evaluated of their impact on system oscillatory stability based on voltage profiles. The dependent load characteristic was found that EV load was the high impact to power system when compared to conventional PQ loads in single machine infinite bus (SMIB) [2]. The EV load from constant power without considering the voltage profile of EV charging system. The EV charging system cannot provide the accurate information about behavior of charging system during charging process. Therefore EV load is developed based on ZIP load that can be used to analyze the behavior of EV charger and integrated to a power grid and determined the impacts of EV charging load [3]. A model of new plug-in electric vehicles (PEVs) for power flow studies based on voltage source converter (VSC) and PEV active power from instantaneous state of charge and discharge is proposed in [4]. The PEVs were effected to voltage profile and can improved voltage stability on power system. The experimental results on laboratory scale measured and derived the model of new load subject to the large voltage

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change and their effect on voltage stability studies based on static load model [5]. The queuing theory based in a PQ bus with stochastic characteristics of electric vehicles demand system were closed formulas for real and reactive power of the EV demand system from charging time that can be found in [6]. The bidirectional four quadrant of battery charger operation were controlled reactive power compensation, voltage regulation and peak shaving that to allow bidirectional power transfer is proposed in [7]. The Monte Carlo simulation was generated probability parameter for computing daily load of EV under behavior of charging system. The load models of EV consist of spatial and temporal characteristics, the load model under autonomous charging mode, the load model under coordinated charging and discharging mode were developed in multi objective function based on Particle Swarm Optimization (PSO) that EV coordinated charging and discharging mode had better operated economics than the autonomous charging mode [8].

The rest of this paper is organized as follows: Section 2 proposes the EV load modeling for power flow analysis that consists of EV load modeling, Newton Raphson power flow with EV load, continuation power flow and load voltage deviation. User define modeling and interface is presents in Section 3. Section 4 shows the simulation results based on IEEE 14 bus test system. Finally, the conclusion and discussion are given in Section 5.

II. EV LOAD MODELING FOR POWER FLOW ANALYSIS

A. EV load modeling

Fig. 2 showed the voltage dependent load characteristics that was represented of EV load model based on the group of Plug-in Electric Vehicles (PEVs) charger in power grid connection. The dc-dc convertor stage is to achieve the required charge current suitable for different state of charge condition and cell temperatures control of battery. Generally the EV load model can be shown in [1],[2] from equation (1) and in practical can be done using in equation (2) and (3), respectively. The EV load model are required parameters for completely load that came from experimental results by testing the EV fast charger.

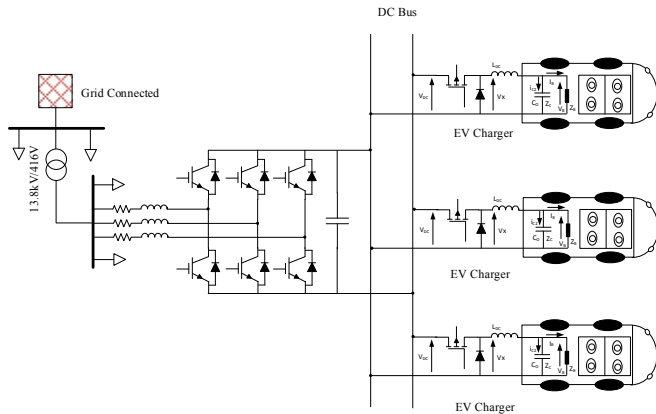


Fig.2 The ac-dc converter at the grid interface and dc-dc converter of a PEVs charger [9].

$$\frac{P}{P_0} = b + a \left(\frac{V}{V_0} \right)^\alpha \quad (1)$$

Therefore;

$$P = b + a \left(\frac{V}{V_0} \right)^\alpha \quad (2)$$

$$Q = P \times \tan(\theta) \quad (3)$$

where V and V_0 are the rated voltage and nominal voltage. The θ indicates the power factor of the connected load, while a, b, α are the constant power, voltage dependent power and exponential component of load, respectively. The P_0 refers to the real power consumption of load at voltage V_0 . So that the real and reactive power of EV load are given by P and Q , respectively.

B. Newton Raphson Power Flow with EV load

The EV load identified voltage dependent load (VDL) characteristics. Therefore, accommodating it in Newton-Raphson power flow algorithm requires certain modifications to the algorithm. The required modifications to the real and reactive power flow equations of the algorithm are shown in equation (4) and (5) [10].

$$P_{Gi} - P_{Di}(v) = v_i \sum_{k=1}^{N_B-1} v_j \left[G_{ij} \cos \delta_{ij} + B_{ij} \sin \delta_{ij} \right] \quad (4)$$

$$Q_{Gi} - Q_{Di}(v) = v_i \sum_{k=1}^{N_{PQ}} v_j \left[G_{ij} \sin \delta_{ij} + B_{ij} \cos \delta_{ij} \right] \quad (5)$$

where δ_{ij} is the bus voltage angle between buses i and j . The total number of buses and number of load buses are represented as N_B and N_{PQ} , respectively. The injection real and reactive power at bus i are represented by, P_{Gi} and Q_{Gi} , respectively. The real and reactive power consumption of the load at bus i are given by P_{Di} and Q_{Di} , respectively. The conductance and susceptance of feeder ij are described by G_{ij} and B_{ij} , respectively. Consequently, the diagonal elements of J_2 and J_4 of standard power flow Jacobian should also be modified. The modification required are shown from equation (4) and (5) in term differential equation into sparse matrix of Newton-Raphson power flow algorithm.

C. Continuation Power Flow Method (CPF)

In order to study the impacts of different loading margin for sizing of PEVs in an electrical power system. Therefore, the CPF method was used to solve the standard power system model from a set of nonlinear differential algebraic equation, as follows: [11]

$$\dot{x} = f(x, y, p) \quad (6)$$

$$0 = g(x, y, p) \quad (7)$$

where $\dot{x}$ are the state variable $x \in \mathbb{R}^n$; y are the algebraic variable $y \in \mathbb{R}^m$; p are the independent variables $p \in \mathbb{R}^l$; f are the differential equations $f: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^l \mapsto \mathbb{R}^n$; and g are the algebraic equations $g: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^l \mapsto \mathbb{R}^m$

The CPF method consists in a predictor step and corrector step. The inserting load parameter λ reformulate the load flow equation. The solving problems were applied parameterization technique and can be adapted real and reactive power from equation (4),(5),(6) and (7), respectively [11],[12].

$$0 = f(x, y, \lambda) \quad (8)$$

$$0 = g(x, y, \lambda) \quad (9)$$

where $\lambda \in \mathbb{R}$ is the loading parameter, which is used to vary base case generator and load powers, P_{G0} , P_{L0} and Q_{L0} respectively, as follows:

$$P_G = (\lambda + \gamma k_G) P_{G0} \quad (10)$$

$$[P_L, Q_L] = \lambda [P_{L0}, Q_{L0}] \quad (11)$$

D. Load Voltage Diaviation(LVD)

The LVD was used to solve the bus voltage deviation that effected from load increase into the electrical power system. The LVD need minimize value of load voltage bus can be described in equation (12) [12].

$$LVD = \sum_k^n \left(\frac{V_k^{ref} - V_k}{V_k^{ref}} \right)^2 \quad (12)$$

where V_{ref} and V_k are voltage reference in normally setting at 1 p.u. and load voltage bus, respectively.

III. USER DEFINE MODELS AND INTERFACE

The EV load modeling was develop from equation (2) and (3). By using User Define Models (UDMs) and interface EV model block, the simulink library was designed in PSAT program. The two parts are required relate by convert all electrical power system model and EV model test system in GUI block to M-file data for preparing the first data. The basis method are describes as below.

A. PSAT program

The PSAT is selected for power system analysis which is open source program running on MATLAB and GNU/Octave based software package. The PSAT can be used to analyze and design the electrical power system for solving the complex problem [11].

B. User Define Models (UDMs)

The UDMs were allowed extending the capability of function model defining by user and help end-user to quick set up their own models. The UDMs of EV load model can be created by means of the GUI depicted in Fig.3 and The EV load model on GUI of Simulink library depicted in Fig.4

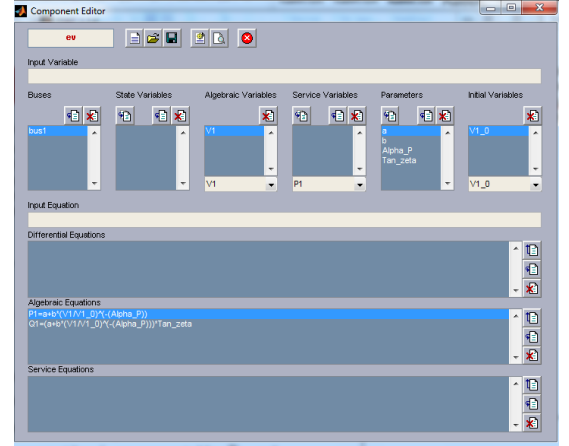


Fig. 3 GUI for User defined models.

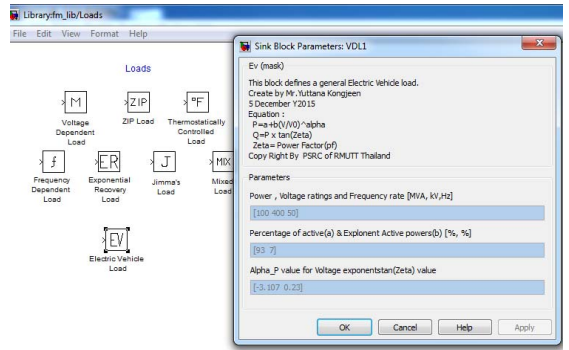


Fig.4 The EV load model on GUI of Simulink in PSAT.

The EV load modeling is created from UDMs function and EV load model on GUI of Simulink library in PSAT must be rerated and linked the parameters. The language programming skill for development EV load model in process of function in PSAT are required.

IV. SIMULATION RESULTS

The IEEE 14 bus system base case used an EV load is connected via 100MVA 13.8kV/416V step-down transformer at bus No.15 as shown in Fig.4. The EV load are represented the group of DC fast charging unit for Plug-in Electric Vehicle in distribution system. To determine the EV load from equation (1) and (2) parameter of a , b , α and power factor are 0.07, 0.96, -3.107 and $pf = 0.96$, respectively. The simulation results showed comparison of voltage profile and PV curve between base case and proposed EV load installation at bus No.15 by vary power rated of EV load

5MVA, 10MVA, 20MVA, 30MVA, 40MVA, 50MVA, 70MVA and 79MVA, respectively.

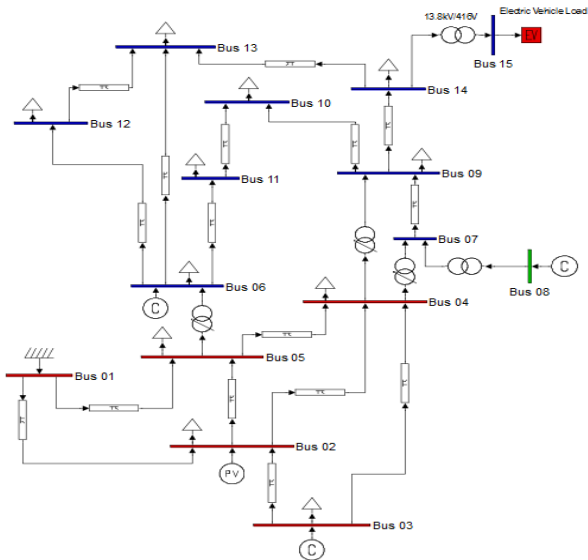


Fig.5 The IEEE 14 bus test system and an EV load fleet via transformer step-down at bus 15.

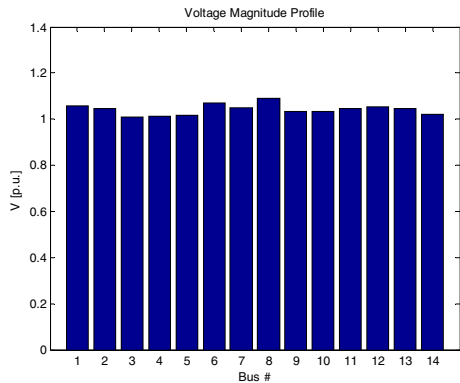


Fig.6 The IEEE 14 bus test system base case

The general load profile base case for the whole load is the depicted in Fig.6. It shows a load voltage of PQ load was installed in the power system. The total power losses and LVD are 0.1358 p.u., 0.2749 p.u. and 0.0313 p.u, respectively.

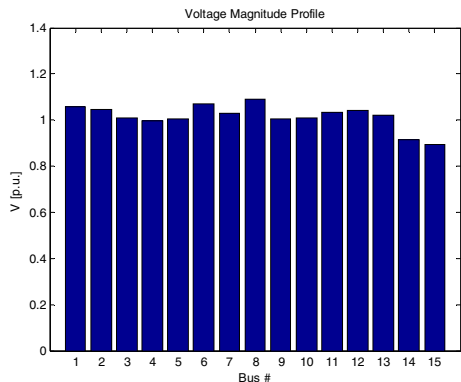


Fig.7 The IEEE 14 bus test system and a EV load 50MVA fleet via transformer step down at bus No.15

Fig.7 showed the load voltage profile with sizing 50MVA of EV load into the power system at bus No.15 that the load voltage profile each bus are changed the total power losses and LVD are 0.2606 p.u., 0.8201 p.u. and 0.0411 p.u, respectively.

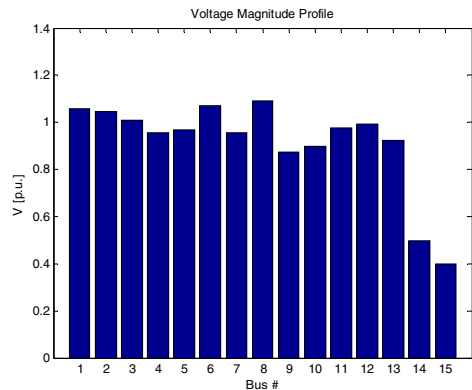


Fig.8 The IEEE 14 bus test system and a EV load 79MVA fleet via transformer step down at bus No.15

Fig.8 showed the high impact of EV load that was effected to load voltage profile with sizing 79 MVA of EV load into the power system at bus No.15. The effecting voltage magnitude was directed at EV load connect to bus No.15 and other bus. The total power losses and LVD are 0.9625 p.u., 3.4328 p.u. and 0.6724 p.u, respectively.

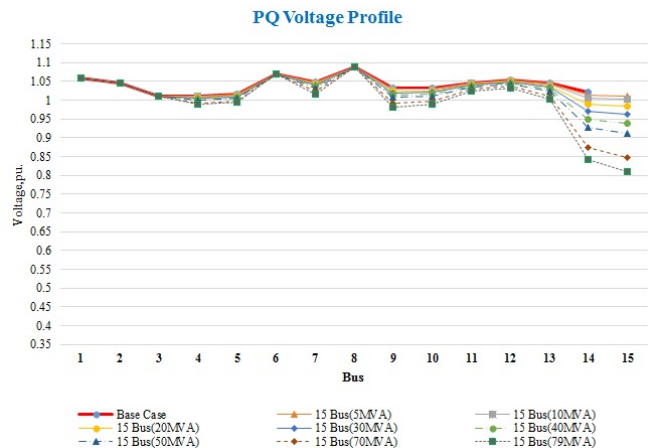


Fig.9 The voltage profile of PQ load when vary the PQ load sizing.

Fig. 9 showed the comparison of voltage profile at each bus that were effected the static voltage stability of the power system when vary sizing of PQ load at $pf = 0.96$ from 5 MVA to 79MVA by installing at bus No.15. The totally results of voltage profile can be evidenced of characteristic the power system with the lowest of voltage as below 0.95 p.u. when vary PQ load more than 30 MVA.

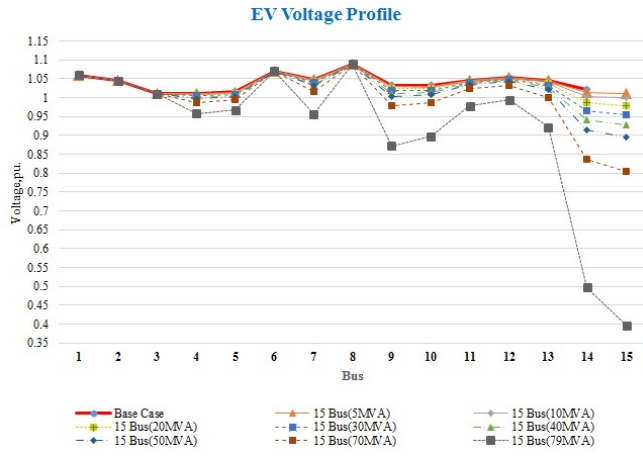


Fig.10 The voltage profile of EV load when vary the EV load sizing.

Fig. 10 showed the comparison of voltage profile at each bus that were effected the static voltage stability of the power system when vary sizing of EV load from 5 MVA to 79MVA by install at bus No.15. The totally results of voltage profile can be evidenced of characteristic the power system with the lowest of voltage as below 0.95 p.u. when vary EV load more than 40 MVA. Therefore, from the voltage profiles found that the characteristics of EV load were effected to the voltage quality more than PQ load. So that optimal of EV load sizing to install in the system was required.

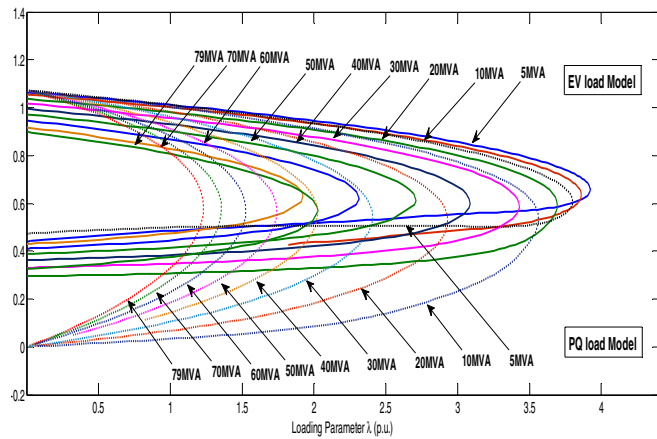


Fig.11 PV curves by vary EV load fleet via transformer step down at bus 15

The simulations above to find out the PV curve by vary sizing of PQ load and EV load at bus No.15. The simulation results were compared characteristic of curve when loading any sizing. The loading margin of PV curve from EV load were changed from 4 p.u. to 1.8 p.u. opposite from direction of sizing EV load, while the loading margin of PV curve from PQ load were changed from 3.6 p.u. to 1.2 p.u.. Therefore, it is necessary to analyze loading margin and capability of power system when carrying any sizing EV load in to power system.

TABLE I. TEST RESULTS

Test Case	Ploss (p.u.)		Qloss (p.u.)		LVD	
	EV	PQ	EV	PQ	EV	PQ
IEEE 14 Bus (Base Case)	0.1358		0.2749		0.0314	
+1 EV/PQ (5MVA)	0.1435	0.1429	0.3092	0.3068	0.0304	0.0305
+1 EV/PQ (10MVA)	0.1519	0.1507	0.3469	0.3414	0.0294	0.0295
+1 EV/PQ (20MVA)	0.1715	0.1684	0.4335	0.4197	0.0284	0.0285
+1 EV/PQ (30MVA)	0.1953	0.1893	0.5378	0.5115	0.0294	0.0289
+1 EV/PQ (40MVA)	0.2244	0.2139	0.6641	0.6188	0.0331	0.0315
+1 EV/PQ (50MVA)	0.2606	0.2430	0.8202	0.7447	0.0411	0.0368
+1 EV/PQ (70MVA)	0.3760	0.3202	1.3061	1.0730	0.0868	0.0611
+1 EV/PQ (79MVA)	0.9625	0.3677	3.4328	1.2713	0.6725	0.0825

The results from Table 1 can be shown that the power loss, reactive loss and LVD from vary sizing of PQ load and EV load in to test system. To increase for sizing of EV load was increased the power loss reactive loss and load voltage deviation of power system more than PQ load. The LVD of system was shown each PQ load and EV load sizing to install in to at bus No.15. However, some sizing of PQ and EV load such as 20MVA can be improved load voltage deviation of power system. Therefore the EV load sizing had been effected total loss in power system and level of voltage deviation when compared with PQ load.

V. CONCLUSION

The modeling of EV load from development based on UDMs in PSAT programing can be used to solve power flow analysis. Therefore, the IEEE 14 bus test system can be used a very simple from GUI and quickly to solve the electrical power system problem from EV load model. In the results of test system show the sizing of EV load model were effected the steady state of voltage stability in term of voltage magnitude profile, total power loss, LVD and loading parameter. Future more, the studying location and sizing to install EV load in power system are necessary to optimize and suitable in the power system for system stability.

REFERENCES

- [1] C.H.Dharamakeerthi ,N.Mithulananthan N. and T.K.Saha, "Impact of electric vehicle fast charging on power system voltage stability," *Electrical Power and Energy Systems*, 2013, pp. 241-249.
- [2] Dharamakeerthi C.H., Mithulananthan N. and Saha T.K., "Impact of Electric Vehicle Load on Power System Oscillatory Stability," *Australasian Universities Power Engineering Conference, AUPEC 2013, Hobart, TAS, Australia 29 September 2013*.
- [3] Ahmed Haidar, Kashem M. Muttaqi, "Behavioral characterization of electric vehicle charging loads in a distribution power grid through modeling of battery chargers," *IEEE Transactions on Industry Application*, VOL. 52, NO. 1, JANUARY/FEBRUARY 2016.

- [4] A.Jimenez,N.Garcia, "Power Flow Modeling and Analysis of Voltage Source Converter-Based Plug-in Electric Vehicles," IEEE Transactions on Power Systems,2011 .
- [5] Rodrigo Garcia-Valle,John G. Vlachogiannis, "Letter to the Editor :Electric Vehicle Demand Model for Load Flow Studies"Electric Power Components and Systems,37:577-582,2009.
- [6] Mithat C. Kisacikonglu,Burak Ozpinneci,Leon M. "EV/PHEV Bidirectional Charger Assessment for V2G Reactive Power Operation"IEEE Transactions on Power Electronics, VOL.28, NO.12, December 2013.
- [7] Haitao Liu,Huaidong Zhuang and Hongbin Wu, "Multi-Objective Dynamic Economic Dispatch of Microgrid System Including Vehicle-to-Grid," Energies,2015.
- [8] Les M. Hajagos,Behnam Danai, "Laboratory measurements and models of modern loads and Their effect on voltage stability studies ," IEEE Transactions on Power Systems,Vol.13 No.2,May 1998.
- [9] Jia Ying Yong, Vigna K.Ramachandaramurthy, Kang Miao, N.Mithulananthan, "Bi-directional electric vehicle fast charging station with novel reactive power compensation for voltage regulation,"Electrical Power and Energy Systems,2015,pp 300-310.
- [10] Dharamakeerthi C.H.,Mithulananthan N., "PEV load and Its impact on static voltage stability,"Plug In Electric Vehicles in Smart Grids Integration techniques,Chapter 8 ,pp.221-248.
- [11] F. Milano, "An Open Source Power System Analysis Toolbox," IEEE Transaction on power systems,March 2005.
- [12] Jittipong Koasungnean,Yuttana Kongjeen,Phanumat Maksaen,Krittidej Buayai, "Reconfiguration and capacitor placement for improve voltage quality and reduce power loss in primary distribution systems," Proceedings of the 7th Conference of Electrical Engineering Network of Rajamangala University of Technology 2015,(EENET2015).